

PREETI CHANDA PATRA

Mumbai, Maharashtra

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Education

Vellore Institute of Technology, Chennai

Sep 2022 – May 2026

B.Tech in Computer Science and Engineering (AI & Robotics)

Relevant Coursework: DSA, DBMS, OS, OOPS, Robotics, AI, ML, Software Engineering, Compiler Design

Technical Skills

Programming: Python, SQL, HTML, CSS, JavaScript, C++ (basics)

LLM Systems: RAG Pipelines, LangChain, AutoGen, Semantic Kernel, Prompt Engineering, Embeddings, Conversational Memory

Machine Learning: Scikit, TensorFlow, PyTorch, Feature Engineering, Model Evaluation

Backend & APIs: FastAPI, REST APIs, LLM API Integration, Tool Calling, Workflow Automation

Data Analysis & Visualization: Pandas, NumPy, Data Cleaning, Statistical Analysis, Matplotlib, Seaborn, Power BI, Tableau, Excel

Computer Vision: OpenCV, Image Processing, Visual Difference Detection

Databases: MySQL, PostgreSQL

Cloud & DevOps: AWS (EC2, S3), Docker, Git

Tools: Jupyter Notebook, VS Code, Postman, Figma, Jira

Fundamentals: OOPs, DBMS, Operating Systems, SDLC, Agile Methodologies,

Internship

Abbott Healthcare Pvt. Ltd.

May 2025 – Jul 2025

Intern Software Developer

Mumbai

- Developed reusable React UI components for enterprise-scale web apps, improving code efficiency and maintainability.
- Executed 150+ unit and integration tests, resolving 25+ bugs to boost system stability and user experience.
- Collaborated with QA and backend teams to ensure accessibility and seamless workflow validation across 150+ pages.

Bellurbis Technologies Pvt. Ltd.

Jul 2025 – Oct 2025

Data Science Intern

Gurugram (Remote)

- Performed data preprocessing and feature engineering using Python, Pandas, and Scikit-learn for model training.
- Built and fine-tuned predictive ML models, improving forecasting accuracy and client insights by 25%.
- Created Power BI and Matplotlib dashboards, delivering actionable intelligence that enhanced stakeholder decisions.

Apmosys Pvt. Ltd.

March 2026 – Present

Data Analyst (AI & ML)

- Built a LangChain-based chatbot with LLM API integration, enabling context-aware responses using prompt engineering.
- Developing an AI visual testing agent for image comparison and UI defect detection using OpenCV, LLMs, VLM and ML techniques.

Projects

AI Video Sort Generator — Next.js, React, TailwindCSS, AI, Clerk, Neon

2025

- Engineered an AI-powered video sorting platform classifying 200+ videos, reducing manual effort by 65%.
- Integrated Clerk authentication with Neon PostgreSQL backend, supporting 10k+ metadata queries.
- Optimized frontend performance using React and TailwindCSS, improving user experience by 40%.

LangChain-Based AI Chatbot with RAG — Python, LangChain, Gemini API, FAISS

2026

- Built a RAG-based chatbot for context-aware Q&A over scraped data using LangChain and Gemini API.
- Developed web scraping pipeline for extraction, cleaning, and chunking with vector storage.
- Implemented conversational memory and prompt engineering to reduce hallucinations and improve accuracy.

Impact of Activation Functions on Vision Transformers — PyTorch, ViT, DINO

2026

- Evaluated GELU, ReLU, SiLU, Tanh, Mish across ViT architectures to analyze performance impact.
- Built DINO-based self-supervised pretraining and fine-tuning pipeline, improving accuracy by 1.5%.
- Achieved 88.98% accuracy on CIFAR datasets, identifying GELU as optimal activation.

End-to-End Data Analytics Project — Banking Domain — MySQL, Power BI

2025

– Developed a data pipeline processing 100k+ records for trend analysis and fraud detection. – Improved reporting accuracy by 30% through data cleaning, transformation, and visualization. – Designed dashboards tracking 8+ KPIs including LDR, churn rate, and segmentation.	
Vendor Performance Data Quality Dashboard — SQL, Python, Power BI	2025
– Built dashboard consolidating 5+ data sources to evaluate vendor performance metrics. – Automated data validation checks using Python, improving accuracy by 25%. – Identified 12% underperforming vendors, enabling data-driven decisions.	
Car Rental Website — React, Node.js, MongoDB, Stripe API	2024
– Developed full-stack car rental platform with real-time booking and availability tracking. – Integrated Stripe API for payments and MongoDB for fleet and booking management. – Implemented authentication and admin dashboard for system control.	
AI Chatbot — Python, OpenAI API, Flask, React	2024
– Built GPT-based chatbot for answering queries from custom knowledge bases. – Developed Flask backend for API handling and conversation management. – Created responsive React UI with chat history and real-time interaction.	
Chat Application — React, Node.js, Socket.io	2024
– Implemented real-time messaging using Socket.io for low-latency communication. – Supported group chats, file sharing, and online presence tracking. – Designed responsive UI with message history and typing indicators.	
Todo List Application — React, LocalStorage	2024
– Built todo app with drag-and-drop reordering and category filtering. – Used LocalStorage for persistent data without backend dependency. – Designed clean UI with dark mode and improved usability.	
Smart Safety Helmet — Arduino, MQ-2, MPU6050	2023
– Developed IoT helmet for gas leakage and accident detection using sensor integration. – Implemented automated alert system to improve safety response time. – Achieved reliable detection using MQ-2 gas and MPU6050 motion sensors.	
Automated Test Tube Holder — Arduino, IR Sensors	2023
– Designed automated test tube handling system using IR sensors to improve lab efficiency. – Enabled precise positioning and reduced manual intervention in workflows. – Built low-cost prototype focused on safety and stability.	

Certifications

- Oracle Certified Foundation Associate (*Jul 2024*)
- Cisco Networking Academy: Networking Basics (*Apr 2024*)
- Spoken Tutorial - IIT Bombay: C, C++, Python (*Apr 2023*)

Positions of Responsibility

Outreach Lead, CloudOps Technical Club <i>Vellore Institute of Technology, Chennai</i>	Aug 2023 – Jul 2025
– Led outreach initiatives for a technical club, engaging with industry partners to secure sponsorships and collaborations. – Organized and executed large-scale hackathons, ensuring smooth event flow from promotion to prize distribution. – Built long-term partnerships with tech companies, enhancing club visibility and increasing sponsorship value.	